

Enhancing Seamless Body Movement Learning with Frame-by-Frame VR Presentation

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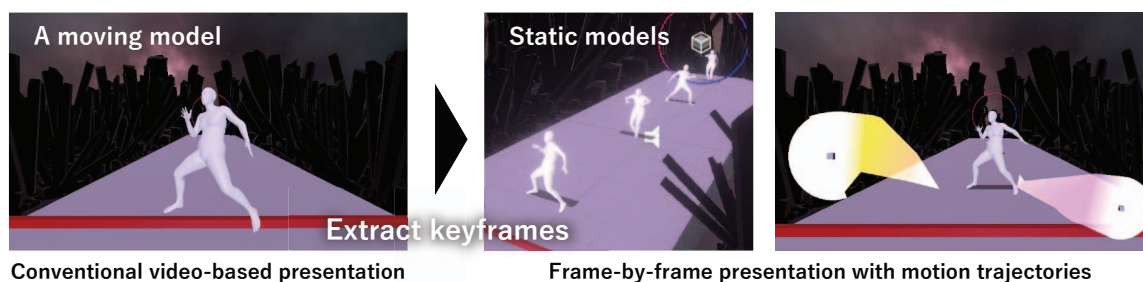


Figure 1: We propose DRF, a virtual reality application for dance training by replaying specific frame images (frame-by-frame presentation) with 3D motion trajectories extracted from dance videos. Compared with video-based presentation methods, learners can more easily associate the poses with the rhythm of the music, potentially leading to an improvement in dance movements.

ABSTRACT

Recent advancements in VR/AR display technology are expected to lead to new ways of learning physical skills through immersive and realistic experiences. However, conventional video-based VR systems often face high cognitive load, especially for complex physical movements such as dance choreography, where multiple intentions and sequential changes must be understood simultaneously. We hypothesized that replaying specific frames extracted from dance videos (frame-by-frame presentation) could reduce this cognitive load and improve the execution of dance techniques. To test this hypothesis, we developed DRF: Dance with Rhythmic Frames, a VR application that combines frame-by-frame presentation with motion trajectories. User study results indicate that with the proposed method, users can improve both choreographic dance technique and rhythm accuracy compared to video-based VR systems. By using frame-by-frame presentation, we aim to solve traditional video-based learning difficulties and support more seamless VR physical training.

Index Terms: Human-centered computing—Human computer interaction (HCI); Human-centered computing—Virtual reality;

1 INTRODUCTION AND BACKGROUND

Recently, advancements in Virtual Reality (VR) and Augmented Reality (AR) display technology are expected to lead to new ways of learning and acquiring physical skills through immersive and realistic experiences [2]. Using motion estimation technology, computers can track the user's movements and provide a realistic training experience. For example, it has been applied to dance training materials using virtual avatars [1, 3, 14], a method for checking the appearance of a ballet performance [11], and ski training materials [9, 10]. These applications address the issues found in 2D video training exercises, which cannot represent depth [13] and are susceptible to occlusions. However, in complex physical movements, such as

dance choreography, which have multiple intentions, the need to simultaneously display and understand numerous changing elements, such as rhythm and complex movements, results in a high cognitive load. Commercial VR dance rhythm games [5–7] reduce this cognitive load by extracting parts of the choreography and presenting them as simple balls or avatars, making complex movements easier to understand. However, it is unclear how effective such simplified presentations are for general body movement learning compared to conventional video learning.

In this paper, we investigate whether replaying specific frames extracted from dance videos (frame-by-frame presentation) can reduce the cognitive load and improve the dance technique execution in actual dance training. To answer that, we propose a VR dance training system DRF: Dance with Rhythmic Frames, combining frame-by-frame presentation with 3D motion trajectory visualization. To investigate the effects of this system in actual dance training, we conducted an experiment with 12 participants using two types of VR systems, one using frame-by-frame presentation (DRF) and the other using conventional video-based presentation. Professional dancers scored the participants' accuracy in choreography and rhythm, and the DRF showed a large effect in learning both of them compared to the video-based system. In addition, through qualitative evaluations of experienced, inexperienced participants, we discussed the feasibility of using the frame-by-frame system for learning body movements.

2 DRF: DANCE WITH RHYTHMIC FRAMES

We present DRF: Dance with Rhythmic Frames, a VR dance training system using frame-by-frame presentation with 3D motion trajectories. It is integrated into a VR environment developed using the cross-platform 3D game engine Unity¹ and displayed with a Meta Quest 2² VR headset.

Figure 2 shows the overview of the system. Users stand in the center of the stage within the VR scene. The 3D human models appear four beats before the timing of the pose from a circle at the edge of the floor, move toward the user at a constant speed, reach

¹Unity, <https://unity.com>

²Meta Quest 2, <https://www.meta.com/jp/quest/products/quest-2/>

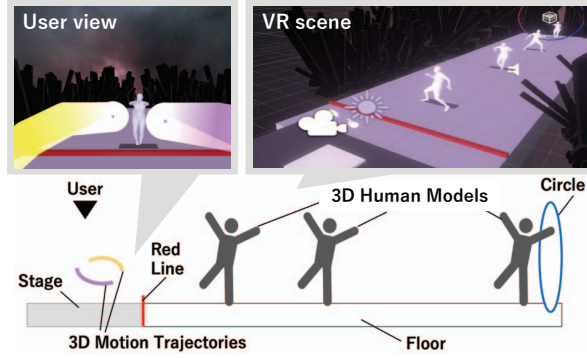


Figure 2: Overview of the VR scene. Users can observe 3D Human Models and 3D Motion Trajectories at the same time. In the user view, the yellow line represents the left wrist trajectory and the pink line represents the right wrist trajectory.

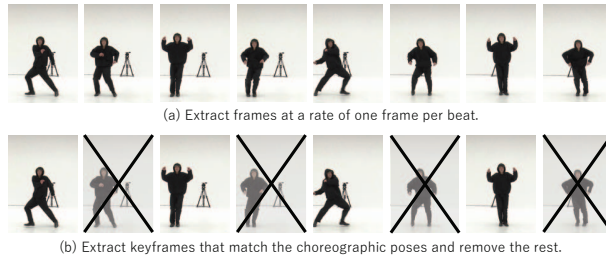


Figure 3: Example of the frame extraction process. First, (a) each frame is automatically extracted from a dance video per beat. Then, (b) manually omit frames that do not match the choreographic key poses.

the red line at the timing of the pose, and then disappear. The 3D models' poses are mirrored, allowing users to learn the dance steps by simply imitating the poses.

Frame-by-frame Presentation of 3D Static Human Models. The keyframe extraction process is as follows. One frame is automatically extracted for each beat using FFmpeg³ as dance poses are typically made in sync with beats (Figure 3 (a)). Then, manually omit frames where the same pose is maintained or between key poses (i.e., frames that are not key poses of the choreography) (Figure 3 (b)).

In this work, we used the BEV [15] platform to infer the poses of a person in the extracted frames. BEV can create a 3D mesh human model from a single RGB image using the SMPL model [12], which is also utilized in a system like [8]. To display the 3D models in accordance with the appropriate rhythm, a rhythm-based MIDI file is created. This MIDI file is loaded into Unity, and the 3D model is presented in sync with the timing of the sound produced.

Visualization of 3D Motion Trajectories. 3D motion trajectories assist in understanding the movement between extracted frames. Users can accurately grasp the depth and direction of movement, thus enabling learning in a manner closer to real-life motion. The method for visualizing 3D motion trajectories is as follows. Frames are first extracted from dance videos at 60 fps using FFmpeg. BEV outputs skeleton information for each frame, including the x, y, and z coordinates of 71 joints. In this system, users can choose to display any of these joints. In our later user study, only the trajectories of both wrists are visualized (Figure 2 User view). The positions of the

³FFmpeg, <https://ffmpeg.org>

selected joints are visualized as white dots and move in accordance with the tempo of the music, which remains visible for 1 second. To help users understand the position of their hands in VR space, the controllers they hold are also visualized. The trajectories are transparent, so the user can always see the 3D models.

3 USER STUDY

We conducted a user study to determine the effects of different presentation methods (video and frame-by-frame) on the participants' learning effects in learning dance steps.

We recruited twelve participants (5 males, 7 females) aged 20-27 years ($M: 22.9$, $SD: 4.08$). 5 participants had dance experience in clubs or dance schools (referred to as "experienced"), while the remaining 7 had dance experience limited to junior high school and high school classes and group dance performances at events such as school festivals (referred to as "inexperienced"). In order to ensure that individual dance experiences did not influence the results of the experiment, we carefully balanced the experienced and inexperienced participants during the study. Regarding their previous VR experience, all participants reported that they had "little to none" or "none at all."

3.1 Dance Movements for Learning

Participants learned two basic dance steps (Step A, Step B) from the AIST Dance Video Database [16] in the Middle Hip-hop genre. These were trimmed to a length of 8 beats, with a quarter note as one beat. All videos were changed to a speed of 3/8 of the original tempo (22.5 fps), as this speed was deemed appropriate for beginners based on preliminary experiments with an inexperienced participant.

3.2 Learning Method

Participants learned two steps using two learning methods, video-based and frame-by-frame. Figure 4 shows the VR scenes for the two learning methods described below.

- Video-based: A moving 3D human model performing dance movements is presented at a fixed position in front of the participant. The model is created using BEV from each reference video and moves smoothly as if the person in the reference video is dancing in front of the participant.
- Frame-by-frame (DRF): Static 3D human models are presented in front of the participant. The models are created from extracted keyframes. The extracted keyframes for each step are shown in Figure 5.

3.3 Experimental Procedure

First, participants wore an HMD (Meta Quest 2) and practiced the dance steps 6 times using the learning method, as our preliminary experiments with one inexperienced participant suggested that inexperienced participants need to practice the steps six times to learn and remember the movements. Participants learned one step using video-based methods and the other step using the frame-by-frame method. The order in which the participants used the two learning methods and the order in which they learned the two steps were balanced well. After participants practiced the step, they removed the HMD and performed a test of the learned step while only listening to the music. All experiments were recorded in front of the participants for later evaluation. To consider mental load, no one observed the participants during the experiment.

3.4 Evaluation Methods

3.4.1 Motion Performance Evaluation

Three professional dancers, with over seven years of dance experience and teaching, rated participants' accuracy in choreography

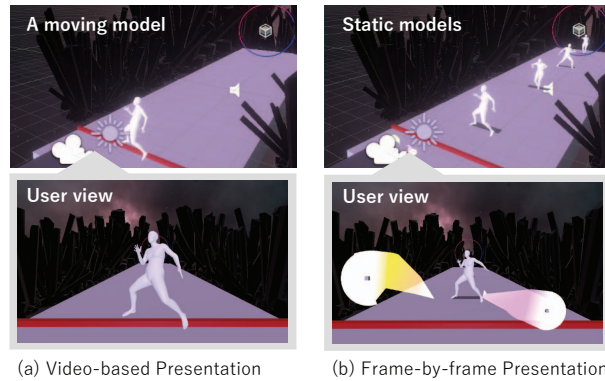


Figure 4: The two scenes have a similar overall design but differ in the way 3D human models are presented.

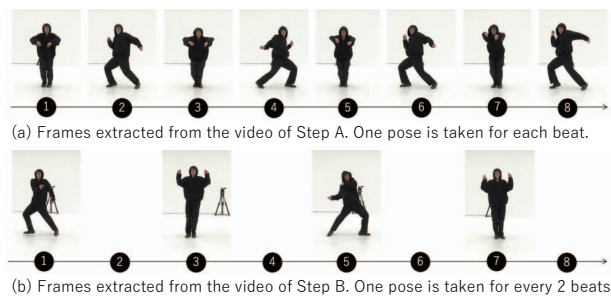


Figure 5: Keyframes from beat 1 to beat 8 of the two dance steps learned in the user study.

and rhythm on a 7-point scale. They evaluated each participant's dance movements against the original reference video, which was rated as 7, using their judgment criteria. Evaluations were conducted without knowledge of which system the participants had used for learning. We created an online form for sharing the dance videos and evaluations.

3.4.2 Qualitative User Evaluation

Participants rated the two learning methods on three aspects: (i) Comprehensibility, (ii) Memorability, and (iii) Usability, on a 7-point scale where 1 strongly prefers video-based methods, and 7 strongly prefers the frame-by-frame method (DRF).

4 RESULTS

4.1 Professional Dancer Evaluation

Figure 6 shows the average scores for the accuracy of choreography and rhythm as evaluated by the three professional dancers, with error bars indicating standard errors. We compare the results for the accuracy of choreography and rhythm trained with the two learning methods, using the Cohen d effect size measure [4]. The effect sizes between the two learning methods among the 12 participants showed a large effect for both choreography and rhythm conditions ($d > 0.80$). Next, we compared the effect size for each of the experienced and inexperienced participants separately, and both showed a large effect in both conditions ($d > 0.80$). This shows that professional dancers tended to rate steps learned with the frame-by-frame method higher for both experienced and inexperienced participants.

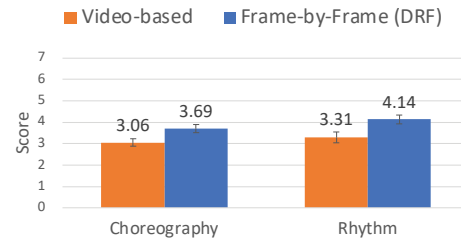


Figure 6: Average scores for choreography and rhythm accuracy as evaluated by the three professional dancers.

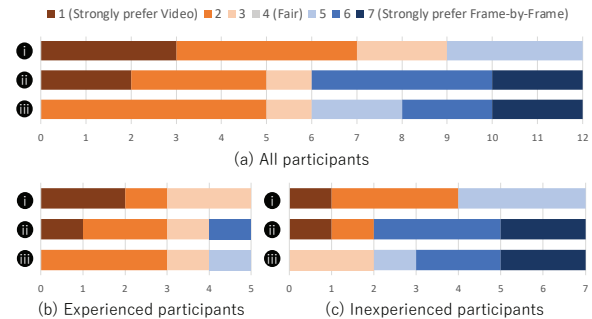


Figure 7: Questionnaire results about the level of (i) Comprehensibility, (ii) Memorability, (iii) Usability.

4.2 Questionnaire of the User Preferences

Figure 7 (a) shows results for all participants. For (i) Comprehensibility, 9 out of 12 participants preferred the video-based method ($M: 2.66$, $SD: 1.70$). This suggests that the video-based method is easier to understand for physical movements. For (ii) Memorability ($M: 4.08$, $SD: 1.49$) and (iii) Usability ($M: 4.08$, $SD: 1.70$), 6 participants preferred video, and another 6 participants preferred frame-by-frame, suggesting that video and frame-by-frame are equivalent in terms of ease of memorization and use.

We next review in detail the results of the experienced and inexperienced questionnaires separately. Figure 7 (b) shows results for the 5 experienced participants, all of whom preferred the video for (i) Comprehensibility ($M: 2.00$, $SD: 0.95$), and also 4 out of 5 preferred the video for (ii) Memorability ($M: 2.80$, $SD: 0.75$) and (iii) Usability ($M: 2.80$, $SD: 1.11$). Figure 7 (c) shows results for the 7 inexperienced participants, with 4 out of 7 preferred the video-based method for (i) Comprehensibility ($M: 3.14$, $SD: 1.41$), and 5 out of 7 preferred frame-by-frame for (ii) Memorability ($M: 5.00$, $SD: 0.75$) and (iii) Usability ($M: 5.28$, $SD: 1.11$). This indicates that experienced participants tend to prefer the video-based method, while inexperienced participants prefer the frame-by-frame method.

5 DISCUSSIONS AND LIMITATIONS

Impact of Frame-by-Frame on VR Learning in Dance. This study suggests the possibility that the frame-by-frame method reduces the cognitive load of understanding complex dance movements. In the user study, professional dancers rated the movements of participants who learned through frame-by-frame more highly in terms of choreographic accuracy, indicating its effects on movement acquisition. This could be due to the frame-by-frame method reducing the amount of information by extracting still images, thus making it easier to understand and remember dance steps. Additionally, professional dancers also highly rated the rhythm accuracy of participants trained with the frame-by-frame method, suggesting that the

rhythm-based frame-by-frame presentation emphasized the timing for dance poses by reducing information overload. This implies a potential reduction in cognitive load for rhythm comprehension compared to video learning. However, as the questionnaire results showed higher ratings for the comprehensibility of movements in video-based method, further investigation from a cognitive science perspective is needed to better understand these factors. Future research should also explore the specific types of movement acquisition suited for frame-by-frame presentation, such as its applicability to other dance genres or sports. Furthermore, in this experiment, we cannot distinguish between the effects of the motion trajectories and those of the frame-by-frame presentation. We plan to conduct future experiments comparing the frame-by-frame presentation without motion trajectory.

System Preferences Based on User Experience Level. Previous studies have primarily focused on either inexperienced or experienced participants, leading to limited effects. This study, however, involved both inexperienced and experienced participants, revealing that user evaluations differ depending on the dancer's level of experience. The frame-by-frame system was particularly preferred by inexperienced participants. In contrast, experienced participants tended to prefer the video-based system, possibly due to their familiarity with learning dance through videos [17]. We will conduct further detailed interviews to clarify the differences in system needs according to the learning level.

Enhancing 3D Model Visualization and Keyframe Extraction. In this study, we motion-captured frames from videos to visualize 3D human models and their motion trajectories. Further investigation is needed on the display methods of the 3D models and the design of the motion trajectories, including aspects like color and pattern. Integrating these elements with feedback presentations could further enhance system effectiveness. Currently, the DRF system requires manual extraction of keyframes. Automating this process and making it adaptable to any video could broaden the system's user base. Future work will focus on developing methods for automatic keyframe extraction.

6 CONCLUSION

In this paper, we introduced a VR dance training system, DRF, utilizing frame-by-frame presentation with 3D motion trajectories. This method involves extracting keyframes from videos and presenting them as 3D human models in sync with rhythm. We conducted a user study to compare the learning effects of this method with conventional video-based VR presentations. The results indicated that our method tends to improve both choreographic dance technique and rhythm accuracy compared to video-based VR systems.

In conclusion, the frame-by-frame presentation is expected to have the potential as an interaction method that effectively reduces cognitive load during complex dance learning. DRF supports seamless VR physical training through the frame-by-frame presentation, making it easier for users to focus on each part of the motion, and the 3D motion trajectory enables learning in a way that is close to reality. However, further research is required to explore more effective presentation methods, such as the display of 3D models and motion trajectories, and visualization of feedback. Additionally, investigating the applicability of the frame-by-frame method to the learning of other sports movements is also necessary for future studies.

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